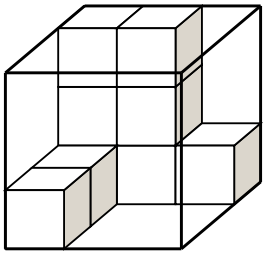
Preview: y2007_q14 (v1)

Space in a box = length × width × height; count hidden cubes; subtract

Part A: Each question is worth 3 points

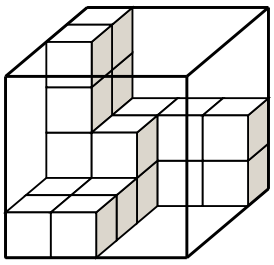
1. Leo has a glass aquarium shaped like a cube, 3 dm long, 3 dm wide and 3 dm tall. Leo has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



At most how many more such cubes can Leo put into the aquarium?

A. 17 ***B.*** 18 ***C.*** 19 ***D.*** 21 ***E.*** 27

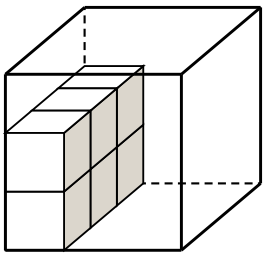
2. Leo has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Leo has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



At most how many more such cubes can Leo put into the aquarium?

A. 44 ***B.*** 43 ***C.*** 42 ***D.*** 41 ***E.*** 20

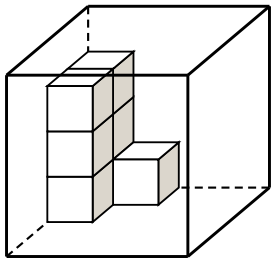
3. Tom puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube.



How many cubes are in the box?

A. 8 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

4. Leo puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube.

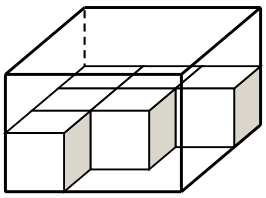


How many cubes are in the box?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 3

Part B: Each question is worth 4 points

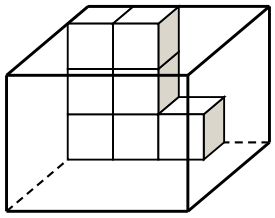
5. Nina's box is 3 cubes long, 3 cubes deep and 2 cubes tall. Some cubes are already inside, as in the picture. Every cube stands on the bottom or on another cube.



How many more cubes fit into the box?

A. 18 ***B.*** 16 ***C.*** 14 ***D.*** 13 ***E.*** 12

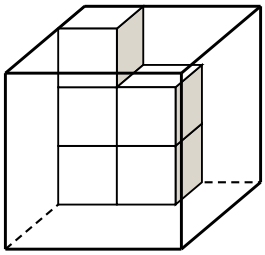
6. Sara's box is 4 cubes long, 4 cubes deep and 3 cubes tall. Some cubes are already inside, as in the picture. Every cube stands on the bottom or on another cube.



How many more cubes fit into the box?

A. 41 ***B.*** 40 ***C.*** 37 ***D.*** 31 ***E.*** 9

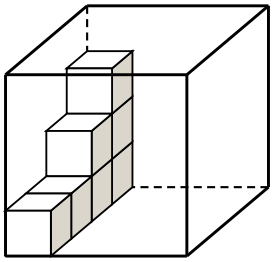
7. Mia puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Now Mia wants to cover the whole bottom of the box with one layer of cubes.



How many more cubes are needed for that?

A. 22 ***B.*** 8 ***C.*** 7 ***D.*** 4 ***E.*** 2

8. Mia puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Now Mia wants to cover the whole bottom of the box with one layer of cubes.

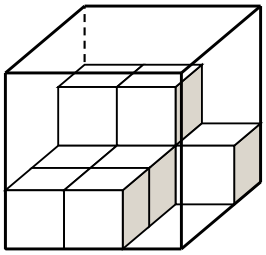


How many more cubes are needed for that?

A. 57 ***B.*** 16 ***C.*** 15 ***D.*** 13 ***E.*** 12

Part C: Each question is worth 5 points

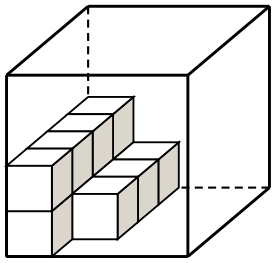
9. Nina puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Nina wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 18 ***B.*** 12 ***C.*** 11 ***D.*** 10 ***E.*** 9

10. Sara puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Sara wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 53 ***B.*** 25 ***C.*** 22 ***D.*** 21 ***E.*** 20

Answer Key

1. **B**	**2. A**	**3. B**	**4. C**	**5. E**	**6. A**
7. **C**	**8. E**	**9. E**	**10. D**		

Solutions

1. B (18) — The aquarium holds 3 × 3 × 3 = 27 cubes. There are already 9 inside (count column by column, also the hidden ones). 27 – 9 = 18. [2007 Q14 · how_many_more]

2. A (44) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 20 inside (count column by column, also the hidden ones). 64 – 20 = 44. [2007 Q14 · how_many_more]

3. B (6) — Count each column from the bottom: 2 + 2 + 2 = 6. [2007 Q14 · count_inside]

4. C (7) — Count each column from the bottom: 3 + 3 + 1 = 7. [2007 Q14 · count_inside]

5. E (12) — The box holds 3 × 3 × 2 = 18 cubes. 6 are inside, so 18 – 6 = 12 more. [2007 Q14 · long_box]

6. A (41) — The box holds 4 × 4 × 3 = 48 cubes. 7 are inside, so 48 – 7 = 41 more. [2007 Q14 · long_box]

7. C (7) — The bottom has 3 × 3 = 9 places. 2 of them already have a column of cubes. 9 – 2 = 7. [2007 Q14 · bottom_layer]

8. E (12) — The bottom has 4 × 4 = 16 places. 4 of them already have a column of cubes. 16 – 4 = 12. [2007 Q14 · bottom_layer]

9. E (9) — Two layers hold 2 × 3 × 3 = 18 cubes. In those two layers there are already 9 cubes (columns count at most 2 each). 18 – 9 = 9. [2007 Q14 · two_layers]

10. D (21) — Two layers hold 2 × 4 × 4 = 32 cubes. In those two layers there are already 11 cubes (columns count at most 2 each). 32 – 11 = 21. [2007 Q14 · two_layers]